Supplementary Material
ARSPI-Net: An Event-Driven Reservoir-Graph Substrate for Affective EEG Perception in Simulated Embodied Control

Supplementary material for the main manuscript. Aggregate, deidentified outputs.

1 Dataset and Privacy Statement

The data are trial-averaged event-related EEG from 211 community-recruited adults (subject 127 excluded for a recording anomaly), giving 633 subject-condition observations balanced across the Negative, Neutral, and Pleasant affective conditions. All evaluation uses subject-grouped cross-validation, so no subject appears in both training and test. Raw EEG, clinical metadata, and subject-level feature artifacts are restricted human-subject research data held in an access-controlled environment; reported outputs are aggregate and deidentified.

2 Feature-Block Definitions

Each observation is decomposed into a spectral baseline and four reservoir-graph evidence streams, each anchored to a named computational object:

- **BandPower**: spectral band-power features (non-neuromorphic baseline).
- E: LIF reservoir spike embedding: per-channel binned spike counts (BSC_6, six windows over $t \in [10, 70]$) compressed by PCA to 64 components per channel.
- D: reservoir dynamical descriptors (variance, autocorrelation, complexity of the reservoir state trajectory).
- T: tPLV graph-topological descriptors (per-electrode strength and clustering of the temporal phase-locking matrix).
- C: structure-function coupling readout $\kappa = \|C\|_F/\sqrt{p\,q}$.

Feature dimensions and reservoir constants are reported in Table 5.

3 Perturbation Protocol

Representation-level perturbations are applied to the test fold only, with the classifier trained on clean training folds (subject-grouped, seeds $\{42, 43, 44, 45, 46\}$):

- **Amplitude noise**: per-column Gaussian noise at $\mathrm{SNR} \in \{20, 10, 5\}$ dB.
- **Channel dropout**: whole-channel zeroing at fraction $\in \{0.1, 0.2, 0.3\}$.
- **Graph edge perturbation**: symmetric tPLV edge dropout at fraction $\in \{0.1, 0.2, 0.3\}$ (affects the T and C streams; non-graph streams are unperturbed).
- **Temporal jitter**: applied in the limited raw-signal diagnostic of the main robustness analysis.

Table 1: Neural mechanisms implemented by ARSPI-Net, each anchored to a concrete computational object and a measured observable.

Neural principle	Implementation	Computational object	Measured observable	Relevance to embodied AI
Event-driven coding	BSC_6 binning	temporal-bin spike-count vector	bin evidence	sparse perceptual observation
Recurrent cortical dynamics	LIF reservoir	reservoir state trajectory	embedding E	nonlinear temporal transformation
Temporal state statistics	descriptor block D	trajectory descriptors	variance, autocorrelation, complexity	dynamical evidence stream
Functional connectivity	tPLV graph	adjacency / graph operator	topology descriptors T	spatial dependency
Structure-function coupling	κ	coupling readout	shuffle-null significance	systems-level coordination
Embodied perception	posterior update + policy	belief state	entropy, success, steps	closed-loop perceptual utility

Table 2: Mechanism ablation under subject-grouped cross-validation. Balanced accuracy (BA) and macro-F1 are reported as mean with a bootstrap 95% interval. Chance BA = 0.333. Each row reports the feature configuration, the neural mechanism it carries, and its feature dimension.

Cfg	Mechanism	Dim	BA	95% CI	macro-F1
A0	spectral band-power (non-neuromorphic baseline)	170	0.485	[0.471, 0.500]	0.483
A1	LIF reservoir spike embedding (E)	2176	0.463	[0.450, 0.476]	0.460
A2	reservoir dynamical descriptors (D)	238	0.432	[0.413, 0.450]	0.429
A3	tPLV graph topology descriptors (T)	68	0.355	[0.343, 0.367]	0.353
A4	structure-function coupling (C)	3	0.368	[0.355, 0.382]	0.352
A5	D+T (dynamics + topology)	306	0.439	[0.425, 0.451]	0.436
A6	E+D (embedding + dynamics)	2414	0.478	[0.465, 0.490]	0.475
A7	E+T (embedding + topology)	2244	0.464	[0.453, 0.476]	0.461
A8	E+D+T (full reservoir-graph substrate)	2482	0.481	[0.468, 0.494]	0.478
A9	E+D+T+C (substrate + coupling readout)	2485	0.481	[0.468, 0.493]	0.478
CTRL_shuffled_label	negative control	2482	0.335	[0.320, 0.351]	0.333
CTRL_shuffled_subject	negative control	2482	0.474	[0.461, 0.487]	0.471

4 Mechanism Ablation Details

The full A0 to A9 mechanism ablation with bootstrap intervals and shuffled-label / shuffled-subject negative controls is shown below.

5 Robustness Tables

The representation-level robustness summary is shown below. Consistent with the main text, the spectral band-power baseline is the most resilient under amplitude noise and channel dropout in the measured regime, the embedding-containing configurations are stable under graph edge perturbation, and the topological stream is the component sensitive to graph-domain perturbation.

Table 3: Representation-level robustness summary: mean balanced accuracy under each perturbation family for representative configurations.

Config	Perturbation	Level (SNR dB / fraction)	BA
A0	amplitude noise	clean	0.485
A0	amplitude noise	20	0.472
A0	amplitude noise	10	0.424
A0	amplitude noise	5	0.395
A0	channel dropout	0.1	0.435
A0	channel dropout	0.2	0.402
A0	channel dropout	0.3	0.408
A1	amplitude noise	clean	0.463
A1	amplitude noise	20	0.460
A1	amplitude noise	10	0.456
A1	amplitude noise	5	0.449
A1	channel dropout	0.1	0.460
A1	channel dropout	0.2	0.454
A1	channel dropout	0.3	0.456
A2	amplitude noise	clean	0.432
A2	amplitude noise	20	0.355
A2	amplitude noise	10	0.349
A2	amplitude noise	5	0.348
A2	channel dropout	0.1	0.336
A2	channel dropout	0.2	0.338
A2	channel dropout	0.3	0.334
A3	amplitude noise	clean	0.355
A3	amplitude noise	20	0.334
A3	amplitude noise	10	0.325
A3	amplitude noise	5	0.327
A3	channel dropout	0.1	0.345
A3	channel dropout	0.2	0.338
A3	channel dropout	0.3	0.342
A3	graph perturbation	0.1	0.351
A3	graph perturbation	0.2	0.342
A3	graph perturbation	0.3	0.331
A8	amplitude noise	clean	0.481
A8	amplitude noise	20	0.401
A8	amplitude noise	10	0.351
A8	amplitude noise	5	0.341
A8	channel dropout	0.1	0.366
A8	channel dropout	0.2	0.334
A8	channel dropout	0.3	0.340
A8	graph perturbation	0.1	0.482
A8	graph perturbation	0.2	0.483
A8	graph perturbation	0.3	0.483

6 Adaptive Evidence Routing

Because the streams fail differently across perturbation regimes, we test whether a router can *select* the stream appropriate to the (measured) signal-quality condition. Each fixed stream is trained on clean training folds and evaluated on perturbed test folds; for every held-out observation we record per-stream predictions, probabilities, and label-free signal-quality observables (max-probability confidence, posterior entropy, scaled feature L_2 norm, and diagonal Mahalanobis distance to the training centroid). Four routers are evaluated:

- **oracle**: selects the stream using test labels; an upper reference bound.
- **perturbation-label**: the regime is known; the best stream is chosen from other folds only (leakage-free).
- **signal-quality**: chooses the stream from label-free observables via a nested subject-grouped meta-classifier.
- **entropy-gated fusion**: fuses the $E{+}D{+}T$ and band-power posteriors when the primary posterior entropy exceeds a training-selected threshold.

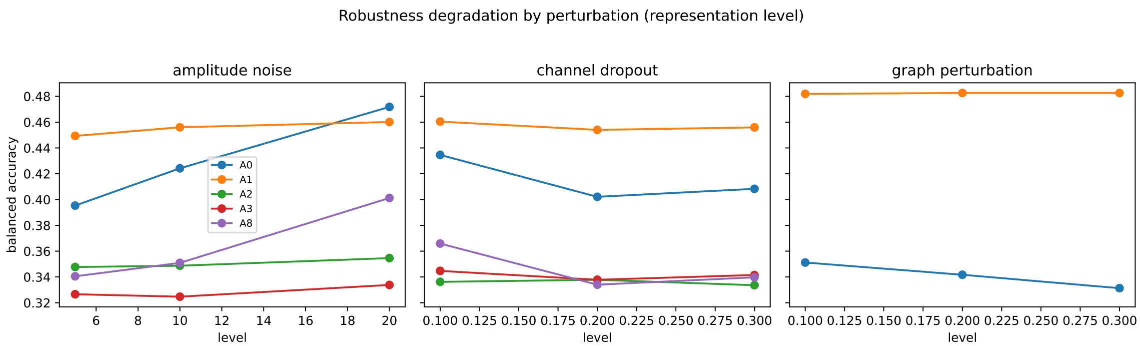

Figure 1: Representation-level robustness degradation by perturbation family.

Table 4: Adaptive evidence routing under perturbation. Balanced accuracy (mean over seeds×folds×regimes). The oracle is a non-deployable upper bound. Routers that use only label-free signal-quality observables are leakage-free under subject-grouped validation.

Router / fixed stream	Balanced acc.	macro-F1
fixed_A0	0.444	0.424
fixed_A1	0.458	0.456
fixed_A2	0.377	0.326
fixed_A3	0.338	0.254
fixed_A5	0.361	0.283
fixed_A6	0.402	0.355
fixed_A7	0.456	0.450
fixed_A8	0.404	0.355
fixed_A9	0.403	0.354
perturbation_label	0.465	0.462
signal_quality	0.438	0.435
entropy_fusion	0.425	
oracle	0.941	0.940

7 Resource and Event-Rate Accounting

We report structural counts, spike-rate measurements, and CPU runtimes for the event-driven substrate. The recurrent layer performs on the order of the measured spike sparsity fraction of the dense-equivalent multiply-accumulate count. *These are computational-resource measurements: event rate, operation counts, and runtime.*

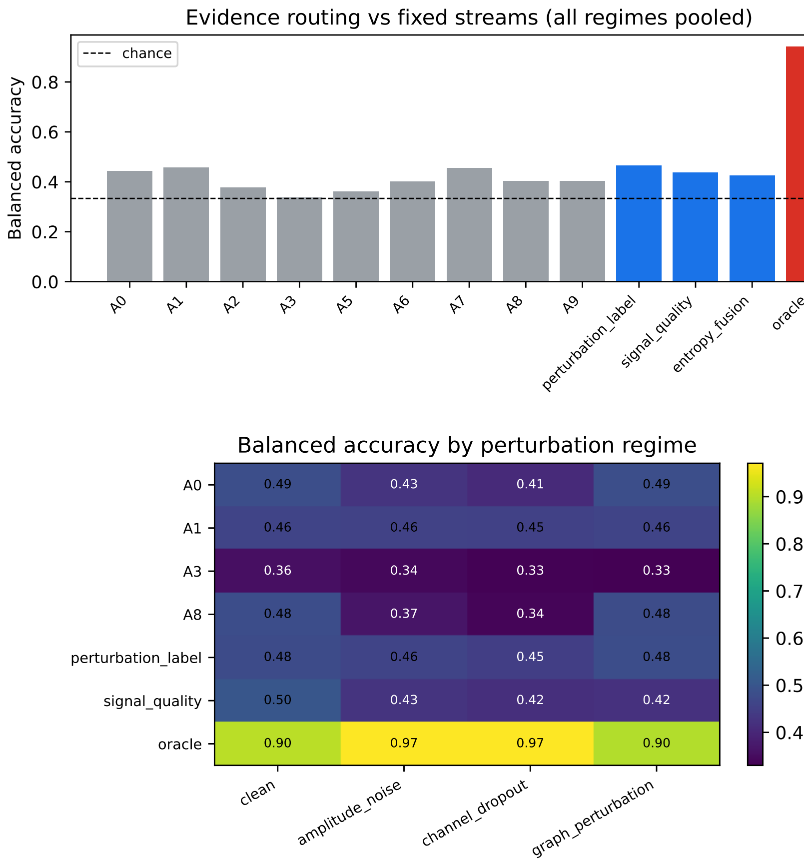

Figure 2: Top: routing vs. fixed streams (all regimes pooled). Bottom: balanced accuracy by perturbation regime. The oracle shows substantial separable headroom across the streams, but the deployable routers (signal-quality, perturbation-label, entropy-gated fusion) remain at or below the best fixed stream under the measured perturbation regime. The routing results quantify the gap between fixed-stream and oracle stream selection, and confirm that the streams are operationally distinct with observable operating regimes.

Table 5: Resource and event-rate accounting for the event-driven substrate. Structural counts, spike-rate measurements, and CPU runtimes; these are not measured hardware-energy results.

Quantity	Value
Reservoir neurons / channel	256
Channels	34
Recurrent weights / channel	65,536
Embedding E dimension	2176
PCA projection weights	3,342,336
Spike sparsity (fraction active)	0.14434
Mean spikes / observation	321612.0
Recurrent ops vs dense	0.1443
Classifier predict (64 obs), s	0.00112

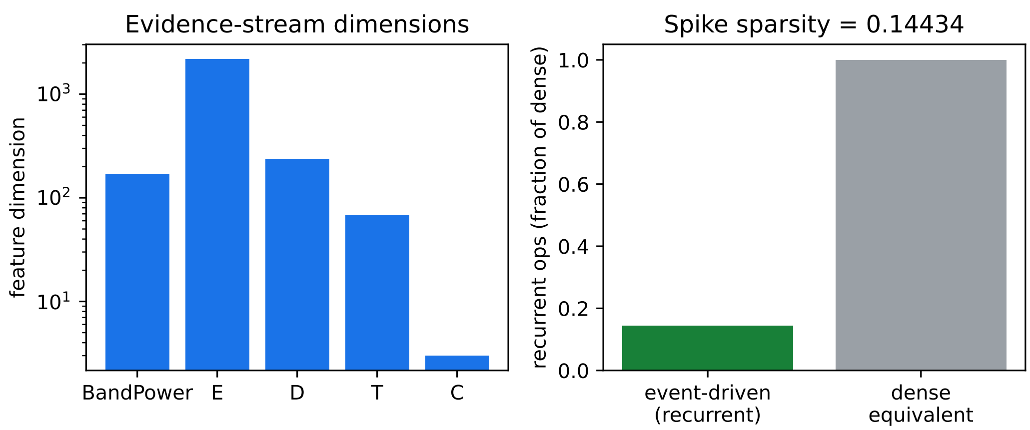

Figure 3: Evidence-stream dimensions and event-driven vs. dense recurrent operation ratio at the measured spike sparsity.